

Command and Control of Drones over 5G

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Abstract—This paper studies the suitability of 5G links for command and control of drones. Using system-level simulations, we evaluate link availability, latency, and block error rate in an urban scenario with drones flying above ground users. The results show that the requirements set by aviation safety authorities for beyond-visual-line-of-sight operations are only partially satisfied: link availability is adequate for basic operations but insufficient for demanding ones such as control-over-camera missions, latency meets most needs but remains too high for direct stick control, and the block error rate is generally too high. These findings suggest that network configurations tailored to aviation safety requirements are needed.

Index Terms—UAV, drones, 5G, cellular networks, system-level simulation, command and control

I. INTRODUCTION

Certain drone applications in fields like agriculture, disaster management, and defense require autonomous beyond-visual-line-of-sight (BVLOS) flights posing technical, regulatory, and safety challenges. One challenge is that wireless command-and-control (C2) must comply with stringent aviation standards, requiring high reliability and low latency. Several technological options exist: Wi-Fi is inexpensive but limited in range and affected by interference, making it unsuitable for BVLOS missions [1]. Proprietary solutions offer performance and range but require upfront investments and infrastructure. Satellite systems offer wide coverage but suffer from higher latency. Finally, cellular networks offer a good balance of performance, coverage, and cost, benefiting from existing infrastructure and pre-acquired frequency bands. New developments like the GSMA Open Gateway [2] enable custom solutions for cellular-connected drones. However, this integration introduces issues like interference and ping-pong handovers due to line-of-sight links of drones to multiple base stations [3], [4].

In this context, where aerial vehicular technology intersects with wireless networks, our research explores the capability of 5G to support BVLOS missions. We study how 5G-connected drones meet the requirements of regulatory bodies, specifically the International Civil Aviation Organization (ICAO) and the European Union Aviation Safety Agency (EASA), thereby bridging the gap between regulatory bodies in uncrewed aerial systems traffic management (UTM) and telecommunications. More specifically, we investigate the C2 link between drones and their base station to ensure at least 99.9% reliability. Our contributions are threefold: First, we select metrics to evaluate the reliability of C2 links from an aviation safety perspective and propose communication metrics to assess those. Second, we conduct a simulation-based evaluation of the reliability of C2 links in BVLOS drone operations. Finally, we

conclude from simulation results that 3GPP link performance expectations are only partly met.

We proceed as follows: Sections II and III review regulatory requirements and related work. Section IV describes the system model and simulation. Sections V and VI present performance metrics and results. Finally, Section VII concludes.

II. REQUIREMENTS

Regulatory approval for critical aerial operations requires proven reliability. Drone operations with BVLOS flights typically fall under the category “specific” in EASA’s classification system [5]. Operators must complete a well-defined workflow called Specific Operation Risk Assessment [6]. As part of this assessment, authorities assign a rating, the *Specific Assurance and Integrity Level* (SAIL), based on the ground risk class, air risk class, and mitigation measures proposed. The SAIL determines the operational safety objectives (OSOs) that the operator must fulfill. Among these, OSO #6 pertains to command, control, and communications (C3). It states that performance and RF behavior of C2 links and any other communication link essential for flight safety must be evaluated. As the requirements for C3 scale with the level of operational risk, the use of cellular-connected drones is an attractive solution to meet the needs of many use cases.

TABLE I
COMMUNICATION REQUIREMENTS FOR DRONE OPERATIONS [7], [8]

No	Operation	Data rate	Latency	Reliability
A	Steer to waypoints	140 bytes/s	1 s	99.9 %
B	Direct stick steering	140 bytes/s	40 ms	99.9 %
C	Automatic flight on UTM	1550 bytes/s	1 s	99.9 %
D	Approaching autonomous navigation infrastructure	4000 bytes/s	140 ms	99.99 %
E	Control over camera (BVLOS)	24 Mbps (720 p)	140 ms	99.99 %

Initiatives by network operators and regulatory bodies aim to ensure the reliability and safety of C2 links for drones. The 3GPP has specified requirements [9] with communication demands in terms of data rate, latency, and reliability summarized in Table I for different operations. Organizations like GSMA advocate the use of cellular networks for drones [10]–[13], noting that the limited availability of BVLOS flights constrains the scalability of commercial applications [13]. They also encourage participation by UTM stakeholders [11].

III. RELATED WORK

Ultra-reliable and low-latency communication over wireless links have been investigated for many years [14], [15]. However, the specific aspects of drones, particularly 5G-connected drones, have only recently been explored. For example, Luo *et al.* [16] evaluate end-to-end delay and compare LTE and 5G at different flight heights. The need for ultra-reliable low-latency communications to enable BVLOS operations is discussed by Ozger *et al.* [17], in which link-level simulations of a drone in a network are performed, evaluating reliability and latency for a piloting message. Morales *et al.* [18] model delays for different protocols suited for drone communications. Depoorter *et al.* [19]–[21] present an evaluation framework for aerial communications, setting the basis for data links and proposing a model for air-to-ground links. Klügel and Schupke [22] expand that work, treating the studied metrics for drones. However, they consider link-level metrics, lack a system-level approach, and focus on one specific mission.

Closely related to the paper at hand is the experimental research on cellular-connected drones by Homayouni *et al.*, who published a series of papers with a common objective [23]–[25]. Their work investigates the capability of today's 4G/5G networks to support drones and evaluates whether 3GPP requirements for link reliability and latency are met. Based on real-world measurements, they identify gaps between standardized demands and observed performance, particularly under certain altitudes and interference conditions. Nevertheless, recent experimental results by Lee *et al.* [26] indicate that 5G-based links can achieve significantly higher reliability than LTE or Wi-Fi-based solutions, even in unlicensed ISM bands.

IV. SYSTEM MODEL AND SIMULATION ENVIRONMENT

Our aim is to assess the suitability of 5G for BVLOS missions and identify limiting factors. BVLOS flights over urban areas have a ground risk class above five, resulting in a SAIL between IV and VI without mitigation measures, which is considered medium-to-high risk in OSO #6 [5]. To perform such flights, the operator must demonstrate that “the C3 link performance is in accordance with standards considered adequate by the competent authority and/or in accordance with means of compliance acceptable to that authority” [5]. We assess via simulation whether 5G satisfies the corresponding requirements in Table I. The system is simulated using a hybrid approach: the environment is a real-world urban district, whereas user locations, traffic patterns, and wireless links follow standard stochastic models.

Environment, ground users, drones: The environment is the western part of the city of Klagenfurt (see Fig. 1). This area contains a university campus, science park, stadium, and a residential area with apartment complexes and family homes. Five base stations (red dots) provide coverage to 80 ground users and 21 drones. Our analysis focuses on outdoor users: pedestrians using voice over IP (VoIP) sessions (yellow icons) or video streaming (purple) and drones are subject to C2 (the drone to be evaluated is shown as a blue dot). The locations are as follows: all ground users are placed uniformly

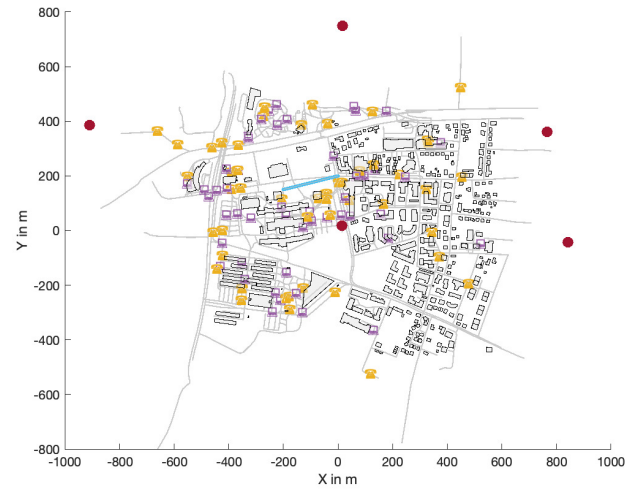


Fig. 1. Environment: Western part of Klagenfurt with base stations (red dots) and ground users with voice (yellow) or video (purple). Drones are not shown.

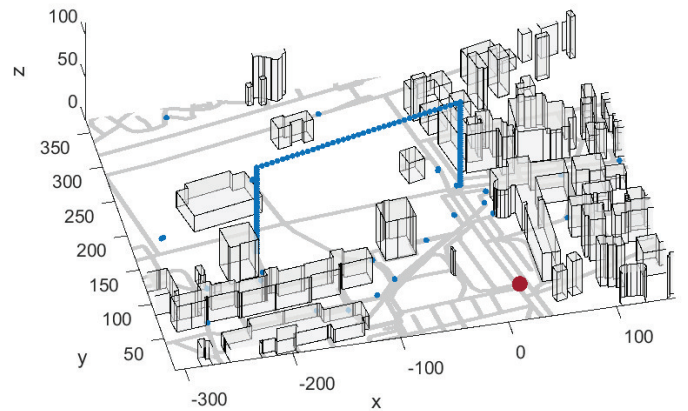


Fig. 2. Flight path of the drone subject to command and control

at random along the streets; drones are positioned uniformly at random within the region of interest at heights ranging from 40 to 120 m. The rationale of not considering other types of ground users, such as vehicular or indoor users, is to limit the complexity of the simulation.

One of the drones is subject to our evaluations. It follows a predefined flight path. Figure 2 shows a typical flight path (blue line) with ascent, cruise, and descent. From a campus point, the drone ascends to 120 m, flies 300 m, and descends. Although this flight path is near a base station, the drone has lines of sight to other base stations when crossing the building threshold and performs handovers (cf. [4]).

Simulation environment: The 5G Vienna System Level Simulator [27] is employed, which supports large-scale network performance evaluation implemented in object-oriented MATLAB. It allows us to modify key network functions, including resource allocation and radio link management. To tailor the simulator to our specific use case, several modifications and extensions are made as described in the following.

Path loss: The default path loss model of the simulator is valid for users below 22.5 m [28]. To support drones, we implement a model from 3GPP recommendations that

TABLE II
ANTENNA: RADIATION POWER PATTERN DEFINITIONS AS PER [28]

Pattern type	Expression	Parameters
Vertical cut	$A''_{\text{dB}}(\theta'', \phi'' = 0^\circ) = -\min \left\{ 12 \left(\frac{\theta'' - 90^\circ}{\theta_{3\text{dB}}} \right)^2, \text{SLA}_V \right\}$	$\theta_{3\text{dB}} = 65^\circ$ $\text{SLA}_V = 30\text{dB}$ $\theta'' \in [0^\circ, 180^\circ]$
Horizontal cut	$A''_{\text{dB}}(\theta = 90^\circ, \phi'') = -\min \left\{ 12 \left(\frac{\phi''}{\phi_{3\text{dB}}} \right)^2, A_{\text{max}} \right\}$	$\phi_{3\text{dB}} = 65^\circ$, $A_{\text{max}} = 30\text{dB}$, $\phi'' \in [-180^\circ, 180^\circ]$
3D pattern	$A''_{\text{dB}}(\theta'', \phi'') = -\min \left\{ -(A''_{\text{dB}}(\theta'', \phi'' = 0^\circ) + A''_{\text{dB}}(\theta'' = 90^\circ, \phi'')), A_{\text{max}} \right\}$	Based on vertical and horizontal cuts

accounts for both line-of-sight (LoS) and non-line-of-sight (NLoS) conditions [9]. The losses depend on the distance d between drone and base station, carrier frequency f_c , and flight height h above ground, with validity for $22.5\text{ m} \leq h \leq 300\text{ m}$:

$$\begin{aligned} \text{PL}_{\text{LoS}} &= 28 + 22 \log d + 20 \log f_c \\ \text{PL}_{\text{NLoS}} &= (46 - 7 \log h) \log d + 20 \log \frac{40\pi f_c}{3} - 17.5 \end{aligned}$$

Antenna patterns: The base station antennas are enhanced to have a mainlobe and two sidelobes using radiation patterns derived from 3GPP specifications [28] (see Table II). Figure 3 illustrates the resulting antenna pattern. The maximum gain of the base station is set to $G_{\text{max}} = 15\text{ dBi}$ [29]. The mainlobe has an elevation of 20 degrees, and the zenith and azimuth angles θ and ϕ are set to 25 degrees. The maximum attenuation is set to -90 dB to achieve a “shadow” effect behind the base stations. Both sidelobes are located above the mainlobe with an elevation of -35 and -70 degrees relative to the mainlobe. The 3 dB angles are decreased by a factor of 0.5 and 0.3 and the level is reduced by 5 and 7 dB, respectively.

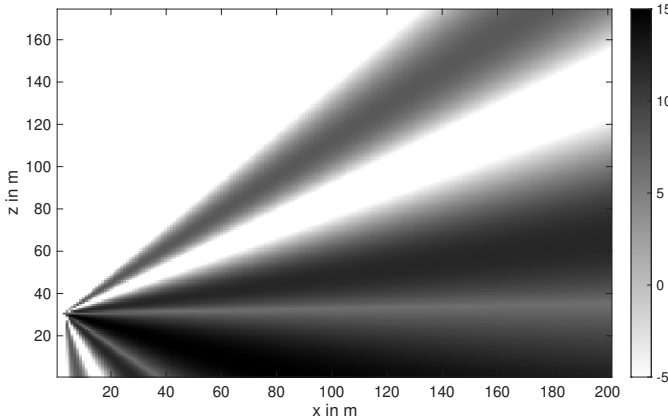


Fig. 3. Antenna: Vertical cut of the gain pattern of an antenna at $z = 30\text{ m}$

Traffic and scheduling: Half of the ground users participate in VoIP sessions; the others perform video streaming. The drones are managed via C2. The packet size and delay constraints of these traffic types are given in Table III. Each base station schedules the downlink traffic using a round-robin or best CQI approach, mapping a resource grid in frequency and time that combines OFDM symbols. Three user grouping schemes are compared: semi-orthogonal user selection (SUS) [30], single path random sampling (SPRS) [31], and

exhaustive search (ES). These determine as to which users can transmit simultaneously in a resource block.

TABLE III
TRAFFIC MODEL PARAMETERS

Parameter	VoIP	Streaming	C2
Packet size (bytes)	320	245	1000
Delay constraint (ms)	60	100	10

Parameters: The system operates with the parameters listed in Table IV. A carrier frequency of 2 GHz is used as many New Radio bands in Europe fall in this range, like Vodafone’s 1.8 GHz (n3) and Deutsche Telekom’s 2.1 GHz (n1).

TABLE IV
SYSTEM-LEVEL PARAMETERS

Parameter	Value
Geometry	Campus Univ. Klagenfurt / Lakeside Park
Area	1.5 km ²
Number of users	80 ground users and 20 + 1 drones
Ground user positions	Random uniform distribution
Drones positions	Random uniform distribution (40–120 m)
Target drone positions	Predefined path
User transmission power	23 dBm
Number of antenna elements	16 (4×4)
Base station transmission power	40 dBm
Pathloss	UMa TR36777 [9]
Frequency	2 GHz
Bandwidth	20 MHz
Number of resource blocks	50
Transmission time interval (TTI)	1 ms
Simulation time	24000 TTI
Feedback type	Closed loop spatial multiplexing
Feedback delay	3 TTI
Scheduler	Best CQI, Round Robin
Grouping	SUS, SPRS, Exhaustive

V. PERFORMANCE METRICS

The UTM framework by the ICAO [32] provides metrics for assessing C2 links. We focus on three of them — availability, latency, and integrity — and, for each, propose a corresponding communication-related metric that has a major impact on it.

Availability: The availability of a link is “the probability that an information exchange between the control station and unmanned aircraft can be initiated when required” [32]. High availability is essential for ensuring reliable remote control of drones. In our setting, we translate this general definition into

the requirement that the effective signal-to-interference-plus-noise ratio (SINR) at the receiver of a wireless transmission must be above a specified threshold SINR_0 . The SINR of a given transmission depends on several factors, including transmit power, antenna gains, path loss, shadowing, and small-scale fading. As a result, different subcarriers generally experience different SINR values. These per-subcarrier values are combined into an effective SINR, which represents the overall link quality between transmitter and receiver. The 5G Vienna System Level Simulator computes this effective SINR using the mutual information effective SINR mapping (MIESM) method. Availability is then represented by the complementary cumulative distribution function (CCDF) of the effective SINR, $P(\text{SINR} > \text{SINR}_0)$, which is the proportion of time for which the effective SINR exceeds the threshold.

Latency: A low communication latency is essential to ensure that drones can be controlled with acceptable delay, enabling responsive maneuvers and real-time actions. The required latency L_0 is “the maximum time permitted for an information exchange to pass one way through the C2 link” [32], representing the time it takes for packets from a base station to a user. Our performance metric is the probability $P(L < L_0)$ that the actual latency L is below L_0 .

Integrity: Another requirement for drone operations is integrity. The level of integrity is “the probability that an information exchange is completed with no undetected errors” [32]. Integrity can be compromised by communication issues and cybersecurity threats. Communication issues include bit and block errors that remain uncorrected and undetected by error control mechanisms. Security threats include authentication and encryption failures. Focusing on the communication issues of integrity, we investigate the block error rate (BLER) of the C2 link. The BLER can be mapped from the effective SINR for a reported channel quality indicator (CQI), i.e., $\text{BLER} = f(\text{SINR}_{\text{eff}}, \text{CQI})$ as a metric for undetected changes.

VI. PERFORMANCE EVALUATION

Our simulations assess the system with respect to the three performance metrics for the five 3GPP flight operations, enabling us to assess whether 5G can support C2 of drones. The results are summarized in Table V. To meet 3GPP requirements, each metric must achieve the values from Table I with a reliability of 99.9% for Operation A, B, and C as well as 99.99% for Operation D and E. If this threshold is missed for 101 ground users and drones, new simulations are performed with fewer ground users and a single drone to find the maximum number meeting the requirements.

A. Availability

The SINR threshold for the C2 link is computed from the capacity $w \log_2(1 + \text{SINR}_0)$ using a bandwidth $w = 20$ MHz. The required data rate depends on the operation type from Table I. For example, Operation D requires 4 kB/s, corresponding to a minimum SINR of -29.5 dB. The most demanding operation, Operation E, requires $\text{SINR} \geq 1.14$ dB.

TABLE V
WHICH OPERATIONS DEFINED BY 3GPP CAN BE FULFILLED WITH CELLULAR TECHNOLOGY FOLLOWING THE DEFINED METRICS?

No	Operation	Availability	Latency	Integrity
A	Steer to waypoints	99.9%✓	99.9%✓	86%△
B	Direct stick	99.9%✓	84%△	86%△
C	Automatic flight on UTM	99.9%✓	99.9%✓	86%△
D	Approaching autonomous navigation infrastructure	99.99%✓	99.99%✓	86%△
E	Control over camera	78%△	99.99%✓	86%△

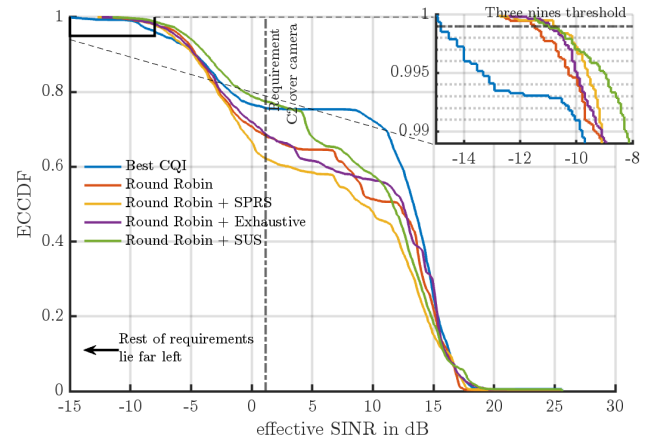


Fig. 4. Link availability: Effective SINR at a drone with different scheduling

Figure 4 shows the empirical CCDF of the effective SINR. In Operations A–D, the requirements are fully met. In Operation E, however, the availability ranges between 62 % (using Round Robin schemes) and 78 % (using Best CQI), which falls short of the criteria for the C2 link. For the upper part of the distribution, during 99.9 % of the time, the data rate is within 112 kB/s (-15 dB) and 275 kB/s (-11 dB). The Best CQI scheduler outperforms the Round Robin schemes.

In conclusion, the simulated 5G system is expected to meet the availability requirements in simple operations but falls short in more demanding ones like control-over-camera.

B. Latency

The most stringent latencies are demanded by Operation B, which requires 40 ms with 99.9 % reliability, and Operations D and E with 140 ms but higher reliability of 99.99 %. Operations A and C can cope with 1 s. Let us explore whether these requirements are met and how scheduling influences the result.

Figure 5 shows the results for all scheduling schemes with 100 ground users. All Round Robin schemes maintain low latency in 98.9 %, with the SPRS grouping yielding the best performance until 99.5 %. Grouping with exhaustive search reaches the 40 ms required by Operation B with a probability of 99.6 %, slightly below the requirement. It meets the requirements of all other operations. The Best CQI scheduler fails to meet the requirement as it prioritizes according to

channel quality, favoring ground users over drones. Given this, 5G is expected to meet the latency demands for typical operations using a suitable scheduler, except for ultra-low latency operations involving safety-critical maneuvers.

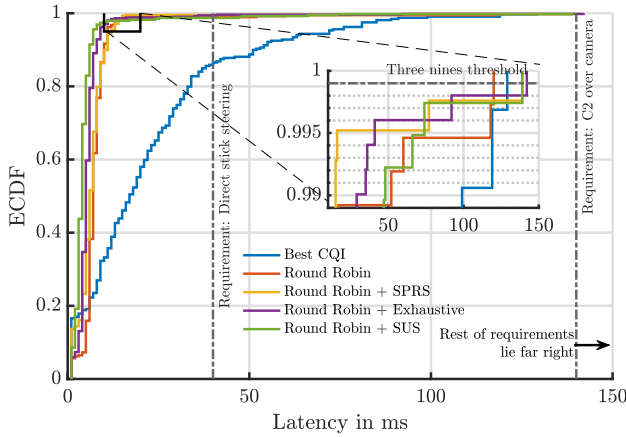


Fig. 5. Latency of the C2 link for different scheduling schemes

We thus examine whether the requirements can be met with fewer users and, if so, determine the maximum number of ground users for which they hold. Figure 6 shows the latency distribution for zero to eleven ground users along with the threshold latency of Operation B (the other operations achieve the required latency). With at most one ground user, the drone's latency stays below 40 ms with high reliability. However, as soon as multiple users are present, the reliability requirement is no longer met (99.6 % for two and three ground users), indicating that the system cannot support direct stick control. If the reliability threshold is relaxed from 99.9 % to 99 %, the requirement is satisfied for up to five users. With eleven users, the reliability at 40 ms further decreases to 96 %.

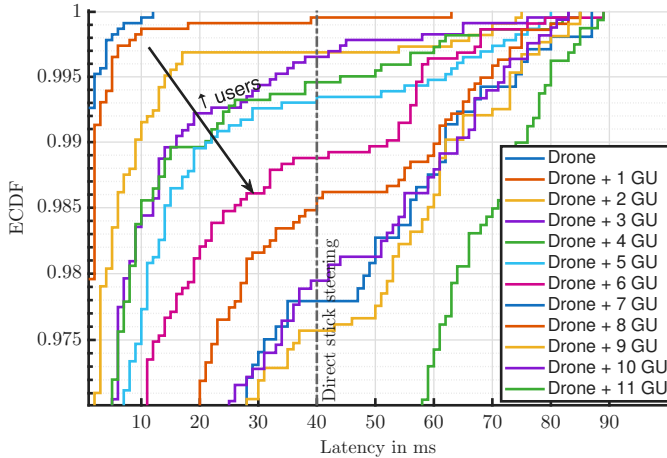


Fig. 6. Latency of the C2 link in the presence of a few ground users (GU)

C. Integrity

The integrity requirement is to achieve a $BLER < 10^{-6}$ with 99.9 % reliability. The simulation results in Table VI show that

none of the schedulers meet this requirement.

TABLE VI
COMMUNICATION INTEGRITY: BLOCK ERROR RATE OF THE C2 LINK

Scheduler	$BLER < 10^{-6}$
BestCQI	84 %
RR Nominal/SPRS/Exhaustive/SUS	83 % / 82 % / 83 % / 86 %

Thus, we again find the maximum number of ground users for which the BLER or a relaxed requirement is met. Figure 7 plots the results over the number of ground users, ranging from zero to eleven. The strict requirement of 99.9 % is met only if the drone is served without any ground users. With a single ground user (and no other drone), the metric drops to 99.85 %, falling just short of the requirement. A more relaxed 99 % threshold is surpassed when three ground users exist. With eleven ground users, the metric degrades to a value below 94 %.

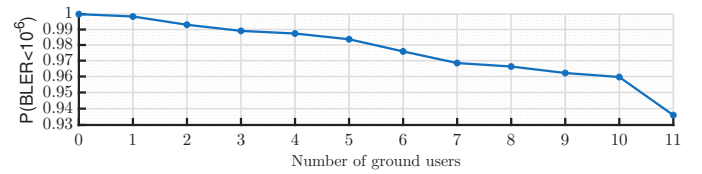


Fig. 7. Communication integrity: Block error rate of the C2 link in the presence of a few ground users

In summary, even when considering only communication errors and excluding security threats, the simulated network is unable to provide acceptable integrity.

VII. CONCLUSIONS AND OUTLOOK

System-level simulations with C2 traffic have been conducted to examine whether 5G is expected to meet the communication requirements for BVLOS drone missions. Safety metrics established by the ICAO and used by aerial traffic management authorities were translated into communication metrics. The results of the simulated system indicate the following: The availability requirement is fulfilled for simple actions but falls short in more demanding operations, such as control-over-camera missions. Furthermore, existing scheduling techniques contribute to coping with the required link performance for the availability of the signal in a command and control scenario. The latency requirements are met in most scenarios by scheduling the users, except for the direct stick control. Finally, the communication integrity requirements in terms of block error rate are generally not met.

Possible future work includes exploring drone-optimized directional antennas, validating the results through real-world experiments, and studying multi-link systems that opportunistically switch between technologies like Wi-Fi, cellular, and satellite systems to improve resilience.

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